Assignment · Build a tree from sequences

Life Sciences · ages 15–18 · Unit 6, Evolution & ecology

In class you built a tree from four short sequences. This time there are six, they are longer, and they belong to six real organisms. Everything you need is on this sheet. No software.

About the sequences. These are simplified 40-base sequences, written for this exercise. They are not downloaded from a database. But they were built so that the differences between them reflect the real evolutionary relationships between these six organisms. If you count correctly, the tree you get is the tree biologists actually accept.

The data

Six organisms. Forty positions each.

```
Y. pestis               ATGCGTACAGACGAGTCCCTTCACGTACGATTCAGGTACC
Y. pseudotuberculosis   ATGCGTACAGACGAGTCCCTTCTGGTACGATTCAGGTACC
Y. enterocolitica       ATGCGTACAGACGAGTCCCTGATCCATAGATTCAGGTACC
E. coli                 ATGCGTACAGACGAAAGAGCGATCGTACCTTTCAGGTACC
Salmonella enterica     ATGCGTACAGACGAAAGAGCGATCGTACGAAGCAGGTACC
Bacillus subtilis       TCAACGGTCTGTCTAAGCCTGATCGTACGATTCAGGTACC
```

Y. pestis causes plague. *Y. pseudotuberculosis* and *Y. enterocolitica* cause gut infections. *E. coli* and *Salmonella* live in intestines. *Bacillus subtilis* lives in soil and is only distantly related to the other five — use it as your outgroup.

Step 1 · Count the mismatches

Take two sequences. Write them one under the other so that position 1 sits above position 1. Go from left to right and count every position where the two letters are different.

Do this for **every pair**. With six sequences there are **15 pairs**. If your table has fewer than 15 numbers, you have missed some.

Write each number into the matrix below. The table is symmetric, so you only need to fill one half — but fill both if it helps you read it.

	Y. pestis	Y. pseudotub.	Y. enteroc.	E. coli	Salmonella	B. subtilis
Y. pestis	—					
Y. pseudotub.		—				
Y. enteroc.			—			
E. coli				—		
Salmonella					—	
B. subtilis						—

Step 3 · How to draw it

Tips on the left, root on the right. Each join is a fork.

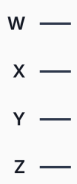
The horizontal position of a fork shows how different the two groups are. A pair that differs by very little gets a fork close to the tips. A group that joins at a large distance gets a fork far out to the right, near the root.

Here is a worked example with different data. Four sequences, W, X, Y and Z, with this matrix:

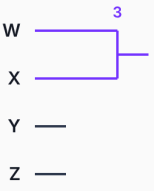
	W	X	Y	Z
W	–	3	9	15
X	3	–	9	15
Y	9	9	–	14
Z	15	15	14	–

Build it one join at a time. Notice how each new fork sits further to the right than the last — that is the distance showing up as a position on the page.

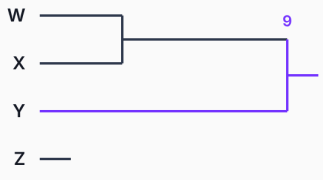
1 Start with the tips in a column.



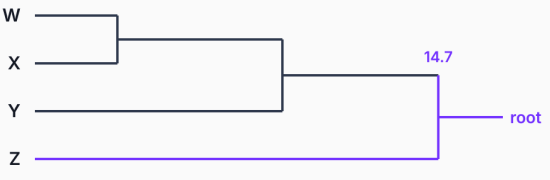
2 Smallest number is 3, between W and X. Their fork sits close to the tips.



3 (W, X) is now one unit, 9 from Y. Y joins at a fork further right.



4 Z joins last, at the largest distance. The root comes out of that final fork.



In panel 3, (W, X) counts as one unit. Its distance to Y is the average of 9 and 9, so 9. Its distance to Z is 15, and Y to Z is 14. The smallest of those three is 9, so Y joins next — and because 9 is bigger than 3, its fork sits further right.

Written in brackets: ((W, X), Y), Z — read it from the inside out. The innermost bracket split most recently.

Three rules that matter

- **Label every fork with the distance at which it formed.** A tree without those numbers only shows the order of splitting, not how far apart anything is.
- **The vertical order of the tips carries no meaning.** You can swap the two branches at any fork and it is still exactly the same tree. (W, X) and (X, W) are identical.
- **A tree with no outgroup has no direction.** Without Z in the example above, you would know W and X are close and Y is further, but you could not say which of them branched off first. The outgroup is what tells you where the bottom is.

What to hand in

1. **The completed matrix.** All 15 numbers.
2. **Your drawn tree.** Hand-drawn is fine. Every fork labelled with its distance, the root marked, the outgroup clearly the last to join.
3. **The tree in bracket notation.**
4. **Answers to the four questions below.** A few sentences each.

Questions

1. Which two organisms are the most similar, and by how many positions out of 40? One of them causes plague and the other causes a mild gut infection. What does your number suggest about how recently they separated?

2. Look at the last eight positions of every sequence. They are identical in all six organisms. Suggest a reason why some positions in a gene almost never change, while others change freely.

3. *Bacillus subtilis* is the outgroup here. Point to the specific numbers in your matrix that show this. Then explain what would happen to your tree if you removed *Bacillus* and worked with the other five only.

4. You built this tree by counting letters, knowing nothing about the organisms. Now compare the tree with what you know about them. Name one grouping that your tree gets right, and explain why counting letters was able to find it.

Checking your own work before you hand it in

- You should have **15 numbers**, not 14 and not 21.
- Your **first join** should be the smallest number anywhere in the matrix.

- Your **outgroup** should have the largest numbers to everybody else. If it does not, recount.
- Some distances in your matrix will be **equal to each other**. That is not a mistake — it is a clue. If two organisms are both the same distance from a third, it usually means those two belong to the same group.